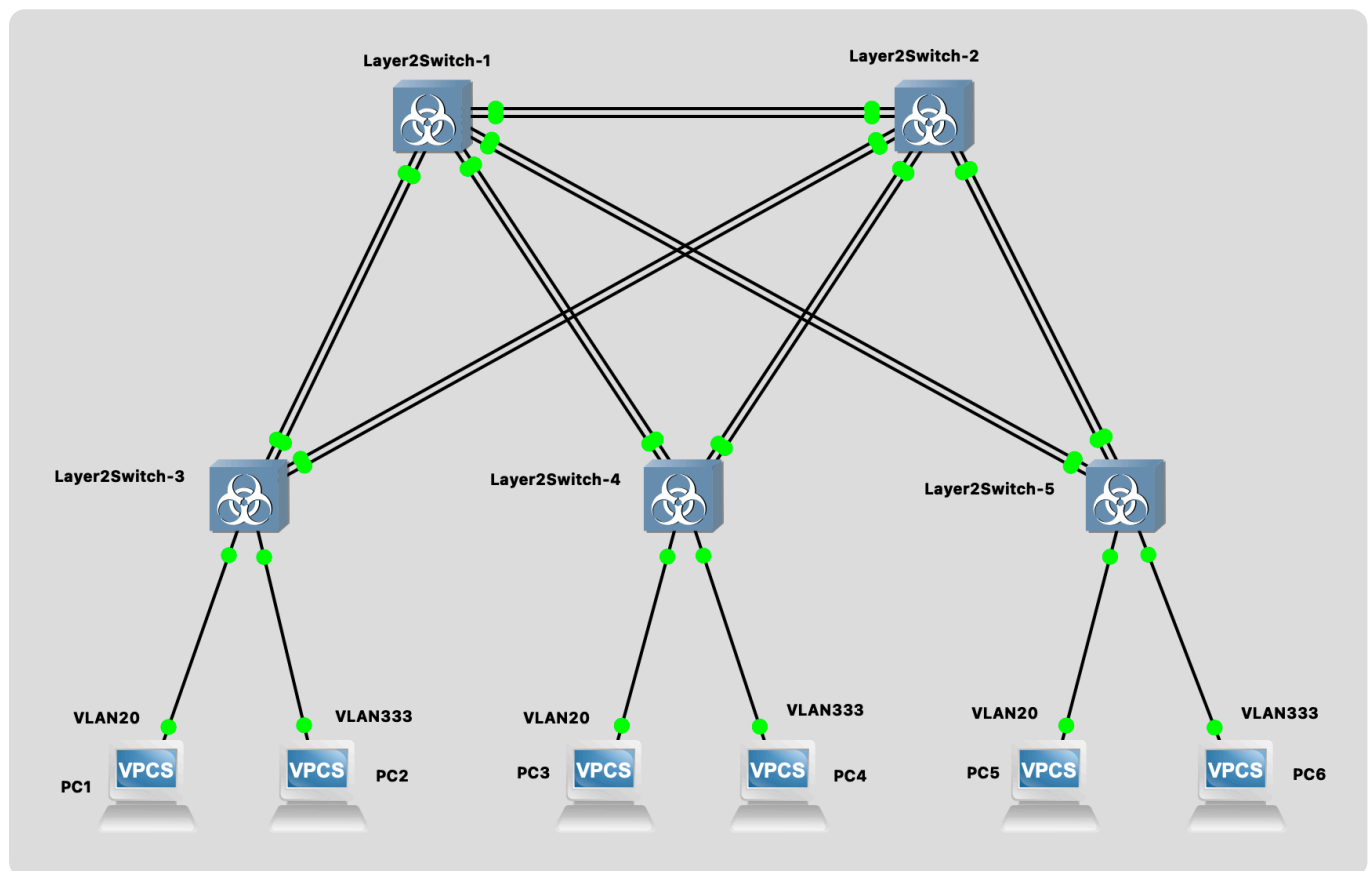


ОТЧЁТ О ВЫПОЛНЕНИИ ЛАБОРАТОРНОЙ РАБОТЫ №3

НАСТРОЙКА ВИРТУАЛЬНОЙ ЛОКАЛЬНОЙ СЕТИ (VLAN)

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Топология



1. Настройка логической топологии IEEE 802.1Q с Native VLAN 333

S1

```
S1>enable
S1#config terminal
S1(config)#vlan 20
S1(config-vlan)#name VLAN20
S1(config-vlan)#exit
S1(config)#vlan 333
```

```

S1(config-vlan)#name VLAN333
S1(config-vlan)#exit

S1(config)#interface gigabitEthernet 0/0
S1(config-if)#switchport trunk encapsulation dot1q
S1(config-if)#switchport mode trunk
S1(config-if)#exit

S1(config)#interface gigabitEthernet 0/1
S1(config-if)#switchport trunk encapsulation dot1q
S1(config-if)#switchport mode trunk
S1(config-if)#exit

S1(config)#interface gigabitEthernet 0/3
S1(config-if)#switchport trunk encapsulation dot1q
S1(config-if)#switchport mode trunk
S1(config-if)#exit

S1(config)#interface range gigabitEthernet 0/0 - 3 , gigabitEthernet 1/0 - 3
S1(config-if-range)#switchport trunk native vlan 333

S1(config)#exit
S1(config)#write

```

```
S1#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	trunk	auto	auto	unknown
Gi1/1		connected	trunk	auto	auto	unknown
Gi1/2		connected	trunk	auto	auto	unknown
Gi1/3		connected	trunk	auto	auto	unknown
Gi2/0		notconnect	1	auto	auto	unknown

```
show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	desirable	n-isl	trunking	333
Gi0/2	on	802.1q	trunking	333
Gi0/3	desirable	n-802.1q	trunking	333
Gi1/0	on	802.1q	trunking	333
Gi1/1	desirable	n-isl	trunking	333
Gi1/2	desirable	n-isl	trunking	333
Gi1/3	desirable	n-isl	trunking	333

S2

```

enable
config t

```

```

vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit

interface gigabitEthernet 0/0
switchport trunk encapsulation dot1q
switchport mode trunk
exit

interface gigabitEthernet 1/2
switchport trunk encapsulation dot1q
switchport mode trunk
exit

conf t
interface range gigabitEthernet 0/1 - 3 , gigabitEthernet 1/0 - 3
switchport trunk native vlan 333

exit
write

```

```
show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi1/2		connected	trunk	auto	auto	unknown

```
show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333
Gi0/1	desirable	n-isl	trunking	333
Gi0/2	desirable	n-isl	trunking	333
Gi0/3	desirable	n-isl	trunking	333
Gi1/0	desirable	n-isl	trunking	333
Gi1/1	desirable	n-isl	trunking	333
Gi1/2	on	802.1q	trunking	333
Gi1/3	desirable	n-isl	trunking	33

S3

```

enable
config t
vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit

```

```

interface gigabitEthernet 0/1
switchport trunk encapsulation dot1q
switchport mode trunk
exit

interface gigabitEthernet 1/0
switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 20
exit

interface gigabitEthernet 1/1
switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 333
exit

conf t
interface range gigabitEthernet 0/0 , gigabitEthernet 0/2 - 3
switchport trunk native vlan 333

exit
write

```

S3#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/1		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-802.1q	trunking	1
Gi0/1	on	802.1q	trunking	333
Gi0/2	desirable	n-isl	trunking	1
Gi0/3	desirable	n-isl	trunking	1

S4

```

enable
config t
vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit

interface gigabitEthernet 0/1
switchport trunk encapsulation dot1q
switchport mode trunk

```

```

exit

interface gigabitEthernet 1/0
switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 20
exit

interface gigabitEthernet 1/1
switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 333
exit

conf t
interface range gigabitEthernet 0/0 , gigabitEthernet 0/2 - 3
switchport trunk native vlan 333

exit
write

```

S3#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/1		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi0/1	desirable	n-isl	trunking	333
Gi0/2	desirable	n-isl	trunking	1
Gi0/3	desirable	n-isl	trunking	1

S5

```

enable
config t
vlan 20
name VLAN20
exit
vlan 333
name VLAN333
exit

interface gigabitEthernet 0/2
switchport trunk encapsulation dot1q
switchport mode trunk
exit

interface gigabitEthernet 1/0

```

```

switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 20
exit

interface gigabitEthernet 1/1
switchport trunk encapsulation dot1q
switchport mode access
switchport access vlan 333
exit

conf t
interface range gigabitEthernet 0/0 - 1 , gigabitEthernet 0/3
switchport trunk native vlan 333

exit
write

```

S3#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/2		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	desirable	n-isl	trunking	1
Gi0/1	desirable	n-isl	trunking	1
Gi0/2	on	802.1q	trunking	333
Gi0/3	desirable	n-isl	trunking	1

2. Проверка доступности между PC в одинаковых VLAN и недоступности в разных

Ping PC1 (vlan 20) → PC2 (vlan 333):

```
ping 192.168.12.1
```

```
No gateway found
```

Ping PC1 (vlan 20) → PC3 (vlan 20):

```
ping 192.168.11.2
```

```
84 bytes from 192.168.11.2 icmp_seq=1 ttl=64 time=18.060 ms
```

```
84 bytes from 192.168.11.2 icmp_seq=2 ttl=64 time=16.458 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=64 time=11.082 ms
```

Ping PC2 (vlan 333) → PC4 (vlan 333):

```
ping 192.168.12.2
```

```
84 bytes from 192.168.12.2 icmp_seq=1 ttl=64 time=14.326 ms
84 bytes from 192.168.12.2 icmp_seq=2 ttl=64 time=12.563 ms
84 bytes from 192.168.12.2 icmp_seq=3 ttl=64 time=23.899 ms
```

Ping PC2 (vlan 333) → PC1 (vlan 20):

```
ping 192.168.11.1
```

```
No gateway found
```

3. Захват пакетов с тегами и без тегов в Wireshark

Capture link: Layer2Switch-1_Ethernet2_to_Layer2Switch-3_Ethernet0

VLAN 20 (tagged)

:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/1/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 Nearest-Customer_	STP	60 Conf. TC + Root = 32768/1/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/20/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/100/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/200/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/300/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/333/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/20/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/100/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/200/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/300/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/333/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/1/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 Nearest-Customer_	STP	60 Conf. TC + Root = 32768/1/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/20/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003
:ba:00:02 PVST+	STP	68 Conf. TC + Root = 32768/100/0c:68:4b:5d:00:00 Cost = 4 Port = 0x8003


```
> Frame 8: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: 0c:68:e0:ba:00:02 (0c:68:e0:ba:00:02), Dst: PVST+ (01:00:0c:cc:cc:cd)
  > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    ... 0000 0001 0100 = ID: 20
    Length: 50
  > Logical-Link Control
  > Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Spanning Tree (0)
    BPDU Type: Configuration (0x00)
    BPDU flags: 0x01, Topology Change
    Root Identifier: 32768 / 20 / 0c:68:4b:5d:00:00
    Root Path Cost: 4
    Bridge Identifier: 32768 / 20 / 0c:68:e0:ba:00:00
    Port identifier: 0x8003
    Message Age: 1
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
    > Originating VLAN (PVID): 20
      Type: Originating VLAN (0x0000)
      Length: 2
      Originating VLAN: 20
```

VLAN 333 (tagged)

:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/1/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	Nearest-Customer...	STP	60	Conf.	TC + Root = 32768/1/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/20/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/100/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/200/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/300/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/333/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/20/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/100/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/200/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/300/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/333/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/1/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	Nearest-Customer...	STP	60	Conf.	TC + Root = 32768/1/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/20/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003
:ba:00:02	PVST+	STP	68	Conf.	TC + Root = 32768/100/0c:68:4b:5d:00:00	Cost = 4	Port = 0x8003

> Frame 7: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0	0000
> Ethernet II, Src: 0c:68:e0:ba:00:02 (0c:68:e0:ba:00:02), Dst: PVST+ (01:00:0c:cc:cc:cd)	0010
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 333	0020
0000 = Priority: Best Effort (default) (0)	0030
...0 = DEI: Ineligible	0040
.... 0001 0100 1101 = ID: 333	
Length: 50	
> Logical-Link Control	
> Spanning Tree Protocol	
Protocol Identifier: Spanning Tree Protocol (0x0000)	
Protocol Version Identifier: Spanning Tree (0)	
BPDU Type: Configuration (0x00)	
> BPDU flags: 0x01, Topology Change	
> Root Identifier: 32768 / 333 / 0c:68:4b:5d:00:00	
Root Path Cost: 4	
> Bridge Identifier: 32768 / 333 / 0c:68:e0:ba:00:00	
Port identifier: 0x8003	
Message Age: 1	
Max Age: 20	
Hello Time: 2	
Forward Delay: 15	
> Originating VLAN (PVID): 333	
Type: Originating VLAN (0x0000)	
Length: 2	
Originating VLAN: 333	

Capture link: Layer2Switch-3_Ethernet4_to_PC1_Ethernet0

VLAN 20 (untagged)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
2	0.330366	0c:76:63:c0:00:04	0c:76:63:c0:00:04	L2L3	60	Reply
3	2.113256	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
4	4.264512	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
5	6.412456	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
6	8.576769	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
7	10.789054	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005
8	11.150161	0c:76:63:c0:00:04	CDP/VTP/DTP/PAgP...	DTP	60	Dynamic Trunk Protocol
9	11.150494	0c:76:63:c0:00:04	CDP/VTP/DTP/PAgP...	DTP	90	Dynamic Trunk Protocol
10	11.174017	0c:76:63:c0:00:04	0c:76:63:c0:00:04	L2L3	60	Reply
11	12.978216	0c:76:63:c0:00:04	Nearest-Customer...	STP	60	Conf. Root = 32768/20/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8005

> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0	0000
> IEEE 802.3 Ethernet	0010
> Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)	0020
> Source: 0c:76:63:c0:00:04 (0c:76:63:c0:00:04)	0030
Length: 38	
Padding: 0000000000000000	
[Stream index: 0]	
> Logical-Link Control	
> Spanning Tree Protocol	
Protocol Identifier: Spanning Tree Protocol (0x0000)	
Protocol Version Identifier: Spanning Tree (0)	
BPDU Type: Configuration (0x00)	
> BPDU flags: 0x00	
> Root Identifier: 32768 / 20 / 0c:68:4b:5d:00:00	
Root Path Cost: 8	
> Bridge Identifier: 32768 / 20 / 0c:76:63:c0:00:00	
Port identifier: 0x8005	
Message Age: 2	
Max Age: 20	
Hello Time: 2	
Forward Delay: 15	

Capture link: Layer2Switch-3_Ethernet5_to_PC2_Ethernet0

VLAN 333 (untagged)

Time	Source	Destination	Protocol	Length	Info
1 0.000000	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
2 0.286952	0c:76:63:c0:00:05	0c:76:63:c0:00:05	LOOP	60	Reply
3 2.181000	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
4 4.378276	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
5 6.546920	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
6 8.733946	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
7 10.891174	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
8 11.189206	0c:76:63:c0:00:05	0c:76:63:c0:00:05	LOOP	60	Reply
9 11.446304	0c:76:63:c0:00:05	CDP/VTP/DTP/PagP...	CDP	420	Device ID: S3 Port ID: GigabitEthernet1/1
10 13.112072	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
11 15.277725	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006
12 15.729034	0c:76:63:c0:00:05	CDP/VTP/DTP/PagP...	DTP	60	Dynamic Trunk Protocol
13 15.729090	0c:76:63:c0:00:05	CDP/VTP/DTP/PagP...	DTP	90	Dynamic Trunk Protocol
14 17.453783	0c:76:63:c0:00:05	Nearest-Customer...	STP	60	Conf. Root = 32768/333/0c:68:4b:5d:00:00 Cost = 8 Port = 0x8006

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

IEEE 802.3 Ethernet

- > Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
- > Source: 0c:76:63:c0:00:05 (0c:76:63:c0:00:05)
- Length: 38
- Padding: 0000000000000000
- [Stream index: 0]

Logical-Link Control

Spanning Tree Protocol

- Protocol Identifier: Spanning Tree Protocol (0x0000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- > BPDU flags: 0x00
- > Root Identifier: 32768 / 333 / 0c:68:4b:5d:00:00
- Root Path Cost: 8
- > Bridge Identifier: 32768 / 333 / 0c:76:63:c0:00:00
- Port identifier: 0x8006
- Message Age: 2
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

0000

0010

0020

0030

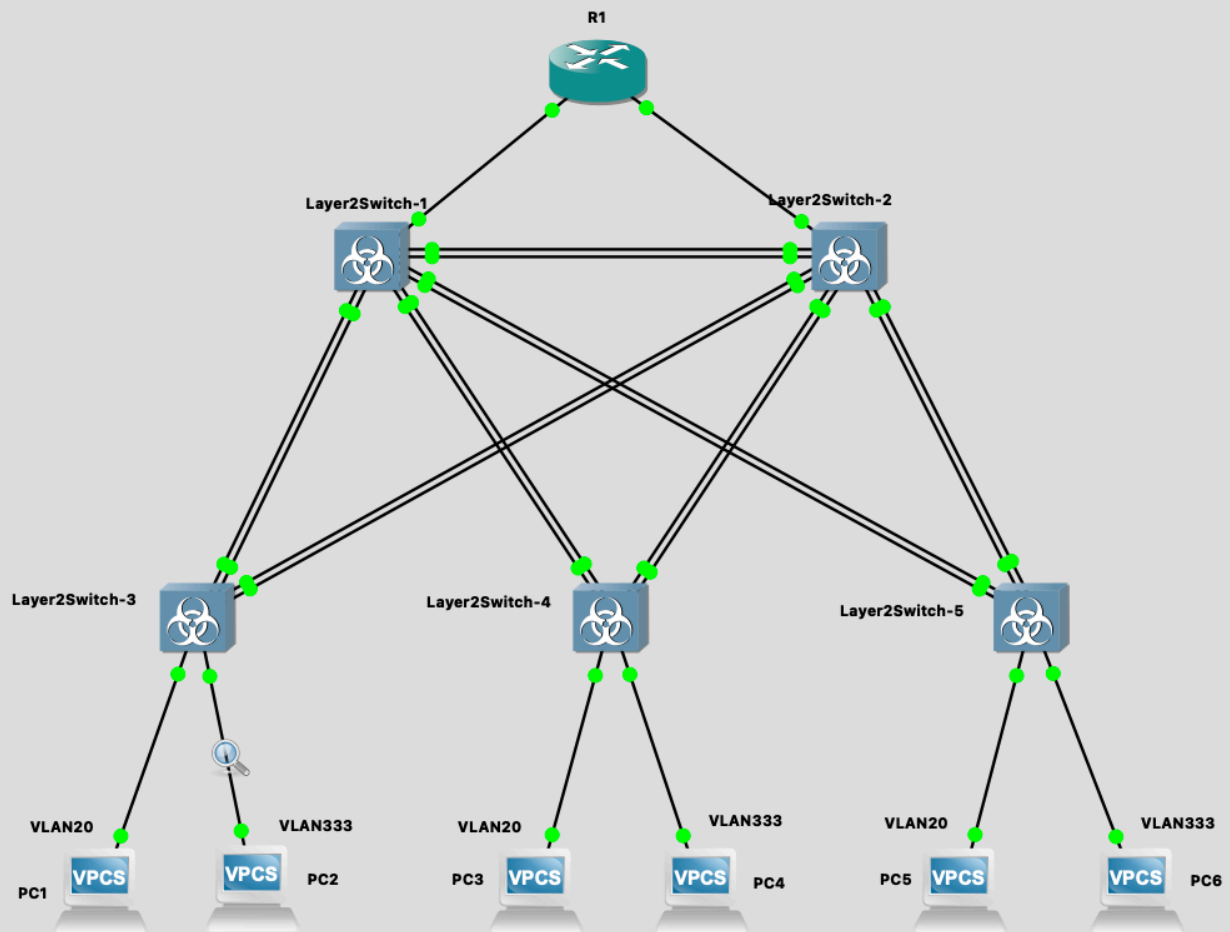
4. Конфигурации устройств

```
enable
sh run
```

Файлы конфигураций сохранены в `./configs`

5. Опциональное задание: маршрутизация между VLAN через роутер

Топология



Поднять интерфейсы R1

```
R1# conf t
R1(config)# interface ethernet 2/0
R1(config-if)# no shutdown
R1(config-if)# end
R1# conf t
R1(config-if)# interface ethernet 2/1
R1(config-if)# no shutdown
R1(config-if)# end
```

```
show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Ethernet2/0	unassigned	YES	unset	up	up
Ethernet2/1	unassigned	YES	unset	up	up

```
show interfaces status
```

```
Interface FastEthernet0/0 is disabled
```

```
Interface FastEthernet1/0 is disabled
```

```
Ethernet2/0
```

Switching path	Pkts In	Chars In	Pkts Out	Chars Out
Processor	40	2460	13	1876
Route cache	0	0	0	0
Total	40	2460	13	1876

Ethernet2/1

Switching path	Pkts In	Chars In	Pkts Out	Chars Out
Processor	6	312	3	728
Route cache	0	0	0	0
Total	6	312	3	728

```
show cdp neighbors
```

Capability Codes: R – Router, T – Trans Bridge, B – Source Route Bridge
S – Switch, H – Host, I – IGMP, r – Repeater

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
S2	Eth 2/1	175	R S I	IOSv	Gig 2/0
S1	Eth 2/0	138	R S I	IOSv	Gig 2/0

Настройка S1 и S2 (access порты для роутера)

```
conf t
interface gigabitEthernet 2/0
switchport mode access
switchport access vlan 20
no shutdown
end
write
```

S1:

```
show interfaces status
```

```
...
Gi2/0                                connected    20          auto    auto unknown
```

S2:

```
show interfaces status
```

```
...
Gi2/0                                connected    333        auto    auto unknown
```

Настройка R1

```
conf t
interface Ethernet2/0
ip address 192.168.11.254 255.255.255.0
no shutdown
interface Ethernet2/1
```

```
ip address 192.168.12.254 255.255.255.0
no shutdown
end
write
```

```
show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Ethernet2/0	192.168.11.254	YES	manual	up	up
Ethernet2/1	192.168.12.254	YES	manual	up	up

Настройка PC1

```
ip 192.168.11.1 255.255.255.0 192.168.12.254
```

Настройка PC2

```
ip 192.168.12.1 255.255.255.0 192.168.12.254
```

Ping PC1 → PC2

```
ping 192.168.12.1
```

```
84 bytes from 192.168.12.1 icmp_seq=1 ttl=63 time=19.553 ms
84 bytes from 192.168.12.1 icmp_seq=2 ttl=63 time=35.031 ms
84 bytes from 192.168.12.1 icmp_seq=3 ttl=63 time=36.131 ms
```

Победа!
